Measurement Dependence Governs Disagreement-Based Reliability in Deterministic MRI Slice Interpolation

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***Abstract*—Deterministic through-plane MRI interpolation gives no indication when a plausible missing slice is inaccurate. Prediction disagreement observes only differences between errors and can miss their shared component. We characterize whether reuse of acquired slice anchors controls this blindness and whether the effect survives architecture changes. The original three-cohort study uses frozen HAT and SwinIR interpolators in 24 fastMRI brain T2 examinations, 115 IXI brain T2 examinations, and 22 NingXia knee T2 series from eight patients. A preregistered fastMRI $\times 4$ extension adds I3Net, an adapted SAINT-style conditional network, and an internal CycleINR-style coordinate model under a clean same-target horizontal-flip shared control and a disjoint phase grid. Disjoint contexts increase observable paired-error energy by 8.0, 13.7, and 22.6 percentage points for I3Net, SAINT, and CycleINR; all examination-bootstrap intervals exclude zero. HAT/SwinIR gains are 17.8/17.6 points, and the equal-family gain remains positive after leaving out any family. Examination-ranking improvement is heterogeneous: strong for CycleINR, small or uncertain for I3Net/HAT/SwinIR, and slightly negative for SAINT. Across factors, diverse architectures still fail on similar examinations (median pairwise error-rank $\rho = .945$–$.981$ on fastMRI), so architecture diversity does not guarantee independent error. The result defines a technical QA consequence and boundary, not calibrated uncertainty or clinical safety: measurement dependence robustly changes local error observability across the evaluated paradigms, while its ranking utility is architecture dependent and agreement cannot certify correctness.**

***Index Terms*—Magnetic resonance imaging, slice interpolation, reliability, uncertainty, disagreement, measurement dependence.**

I. Introduction

ANISOTROPIC MRI volumes often have coarse through-plane resolution. Learned interpolation improves visual continuity and quantitative reconstruction [1], [2], but a frozen deterministic model normally returns one image without stating when the missing anatomy is insufficiently constrained. A deployment-facing reliability signal must identify likely failure without accessing the unavailable dense target.

Ensembles, stochastic sampling, and test-time augmentation estimate uncertainty from output variation [3]–[6]. Their premise is one-sided: disagreement witnesses a difference, while agreement need not imply correctness. Hu *et al.* showed that variability-based uncertainty can miss systematic MRI reconstruction error [7], and MRI ensembles have been evaluated under domain shift [8]. We investigate a complementary, physically auditable source of dependence: two interpolation contexts may reuse the same acquired slices and therefore inherit aligned errors that cancel in their difference.

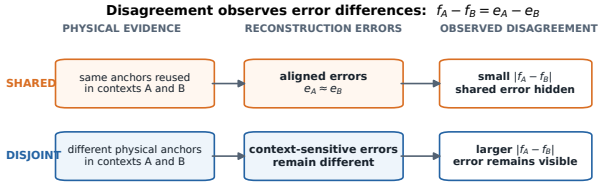


Fig. 1. Consensus blindness. Aligned errors cancel in output disagreement. Measurement-disjoint contexts can reveal more context-sensitive error, but agreement remains ambiguous.

Figure 1 summarizes the hypothesis. We reconstruct the same missing physical plane under contexts that either share no acquired anchors or maximally reuse them. The study does not claim a new correlated-error identity, a new reconstruction backbone, or calibrated uncertainty. Its contribution is a three-cohort medical-imaging characterization of how known measurement dependence changes error observability.

Our contributions are: (i) a target-preserving neural measurement-dependence contrast across three cohorts and a preregistered five-architecture fastMRI falsification; (ii) complete shared/observable error distributions, seven endpoints, and leave-one-family-out analysis; (iii) analytical $\times 2/\times 4/\times 6$, geometry, intensity, strict stable-wrong, and cross-model error-diversity audits; and (iv) examination-level risk–coverage against strong output and acquired-data comparators. Negative boundaries are retained: ranking improvement is not universal, SSIM-error ranking does not generalize, independent output normalization is invalid, and architecture diversity alone does not ensure useful error diversity.

II. Related Work

Classical splines and learned slice synthesis estimate missing samples from smoothness or anatomical priors [1], [2], [9]. Coordinate models such as CycleINR provide a materially different continuous volumetric interpolation paradigm [10]. Leave-one-slice-out evaluation predates this work [11]; we do not claim the operation as novel. Uncertainty in MRI reconstruction has been studied with variational, Bayesian, posterior-sampling, and ensemble methods [5], [6], [8], [12]. These methods learn or sample predictive distributions. We instead ask whether deterministic reliability pairs depend on the same physical evidence and explicitly try to falsify the phenomenon across interpolation paradigms.

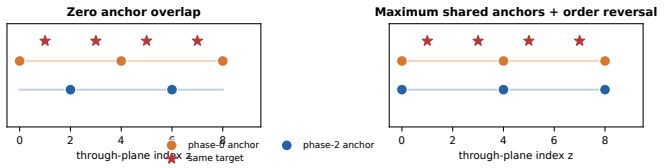


Fig. 2. Target-preserving $\times 4$ design. Stars are evaluated physical planes. Disjoint phase lattices share no anchors; the maximal-shared control reuses the same anchors under reversal.

Ensemble theory distinguishes diversity from accuracy [13], [14]. Our distinction is at the intervention level: target, model, mask, and measurement count remain paired while physical anchor identity changes. This makes dependence observable in a way that arbitrary output transformations do not.

III. Theory and Hypothesis

Let the same target y have deterministic predictions $f_a = y + e_a$ and $f_b = y + e_b$. Their squared disagreement is

$$\|f_a - f_b\|^2 = \|e_a\|^2 + \|e_b\|^2 - 2\langle e_a, e_b\rangle. \tag{1}$$

Positive error alignment suppresses observable disagreement. With $\bar{e} = (e_a + e_b)/2$ and $\delta = (e_a - e_b)/2$,

$$\tfrac{1}{2}(\|e_a\|^2 + \|e_b\|^2) = \|\bar{e}\|^2 + \|\delta\|^2. \tag{2}$$

We call these shared and observable energy. Shared fraction is $q = \|\bar{e}\|^2/(\|\bar{e}\|^2 + \|\delta\|^2)$. This is an algebraic retrospective decomposition, not a calibrated uncertainty partition.

No disagreement score alone identifies error. If $e_a = e_b = c \neq 0$, disagreement is zero while all error is shared. Conversely, by the triangle inequality, large disagreement implies at least one prediction is inaccurate but does not identify which. Our directional hypothesis is therefore scoped: reducing overlap of error-generating measurements should lower error alignment, increase observable fraction, and improve error ranking, but low disagreement cannot certify correctness.

At factor four, phase-0 anchors are $z = 0, 4, 8, \ldots$ and phase-2 anchors are $z = 2, 6, 10, \ldots$. Targets with $z \bmod 4 \in \{1, 3\}$ are missing from both lattices and the lattices are disjoint (Fig. 2). The maximal-shared control reuses phase-0 anchors, reverses their through-plane order, and maps predictions back to physical z. It preserves anchor identity but changes order; the design is a paired intervention with a residual order-equivariance confound, not pure identification of an overlap coefficient.

IV. Materials and Methods

A. Cohorts and Reconstruction

fastMRI brain T2 used a frozen 112/24/24 split; the 24-case test set enters analysis [15], [16]. IXI contributed 115 held-out brain T2 examinations [17]. NingXia contributed 22 knee T2 series from eight patients. Its HAT/SwinIR predictions were out of fold under eight patient-group folds, and inference resamples patients rather than series.

Frozen 2.5-D HAT and SwinIR interpolators generated the original three-cohort contexts; that historical analysis uses the through-plane reversal control and retains its stated order-equivariance limitation. No model was refit for reliability. The architecture extension instead uses a clean in-plane horizontal flip mapped back to the target, preserving physical anchor order and geometry. Complete original 16-plane fastMRI sources provide four phase-2 anchors and therefore validly evaluate I3Net, an adapted locally trained SAINT-style conditional CNN, and an internal CycleINR-style coordinate model. These labels describe architectural lineage rather than exact official checkpoints. Five common missing targets per examination use phase-0 anchors $\{0, 4, 8, 12\}$ and disjoint phase-2 anchors $\{2, 6, 10, 14\}$. An independent factor experiment used analytical interpolation in shallow volumes at $\times 6$.

TABLE I
Cohorts and Experimental Roles

Cohort	Cases	Clusters	Role
fastMRI brain	24	24	Neural overlap, factors, risk
IXI brain	115	115	Neural overlap, factors, risk
NingXia knee	22	8	OOF overlap, factors, risk

The resources deliberately differ in volume depth and intensity distribution. We do not pool raw errors or set a cross-cohort score threshold. Replication means that a within-cohort directional contrast recurs under the same computation. For NingXia, 22 series are observations while eight patients define inference. Multiple planes and architectures remain subordinate to the examination/patient cluster.

B. Physical Matching and Scores

Generated planes were matched using NIfTI affine-derived physical origins. Shapes, orientation codes, affines, plane-normal angles, and distances were audited. Acquired anchors were excluded. For paired predictions a, b, the primary score was mean $|a - b|$ on a fixed foreground mask. Dense targets entered only retrospective acquisition synthesis and evaluation.

Endpoints were MAE, RMSE, MSE, $1 - \mathrm{SSIM}$, gradient MAE, Gaussian low-frequency MAE, and high-frequency residual MAE. The strict stable-wrong definition combined pooled within-dataset/model top-quartile plane MAE and bottom-quartile disagreement. Examination scores were aggregated over matched planes.

For paired errors, total energy was $T = (\|e_a\|^2 + \|e_b\|^2)/2$, shared energy $S = \|\bar{e}\|^2$, and observable energy $O = \|\delta\|^2$. We verified $T = S + O$ before computing fractions. Normalized disagreement divides paired RMSE by total-error scale, making the conditional analysis sensitive to allocation of fixed error energy rather than simply its magnitude.

Three intensity strategies were prespecified. Raw scale preserves the stored computation. Pooled-common normalization applies one robust affine transform to both predictions and target within a case. Independent-output normalization transforms each prediction separately and is a negative control: it can remove amplitude failures differently and therefore does not preserve an absolute paired residual.

The selective-risk endpoint was independently stored mean HAT/SwinIR $\times 4$ examination MAE. Comparators were measurement-disjoint disagreement, maximal-shared disagreement, HAT–SwinIR committee disagreement, and adjacent-anchor MAE. After sorting from low to high score, risk $R(c)$ is retained mean endpoint error at coverage c. Normalized AURC subtracts oracle AURC and divides by reverse-oracle minus oracle AURC; lower is better [18].

C. Statistical Analysis

Spearman correlation measures ordering, not calibration. Primary intervals used 5,000 cluster bootstrap replicates, resampling 24 fastMRI examinations, 115 IXI examinations, or eight NingXia patients with subordinate planes/models retained. Partial rank analyses residualized shared fraction and disagreement against total error and examination indicators. Multi-endpoint confirmation required a positive disjoint-minus-shared direction for a majority of justified endpoints in every eligible cohort; all endpoints are reported. No universal superiority or clinical-safety test is performed.

The endpoint and score are paired by dataset, case, model, and physical plane. Bootstrap samples carry both contexts together, so their difference is recomputed rather than treating correlations as independent. We report point-estimate unanimity but do not count 42 correlated rows as independent discoveries. Complete risk curves complement correlation because the high-risk tail can differ even when overall ranks are similar.

V. Results

A. Original All-Plane Paired Intervention

Before the common three-plane Plan6 analysis, the frozen all-eligible-plane experiment established the target-preserving contrast on fastMRI and NingXia (Fig. 3). On fastMRI, zero-overlap disagreement correlated with phase-0 MAE at $\rho = 0.960$, versus 0.605 for the maximal-shared control, a paired difference of 0.355 (case-clustered 95% CI 0.225–0.557). On NingXia, correlations were 0.891 and 0.699, a difference of 0.193. Correct resampling of eight patient groups gave CI 0.113–0.336.

The effect was positive for each architecture: fastMRI HAT/SwinIR differences were 0.340/0.375 and NingXia differences 0.128/0.258. The endpoint, mask, model, target plane, and number of anchors were paired. Raw disagreement magnitude did not explain the result: on fastMRI, disjoint raw disagreement was only 59.1%/77.2% of maximal-shared disagreement for HAT/SwinIR despite much better error ranking. NingXia had the opposite magnitude ordering. Output diversity magnitude is therefore neither necessary nor sufficient for error observability.

The later three-plane computation uses equal examination coverage and adds IXI plus six endpoints. Its smaller pooled MAE advantages (0.117/0.086/0.185) are not a failed replication of the historical statistic; they reflect a different, prespecified balanced sampling estimand. Both analyses retain the same direction.

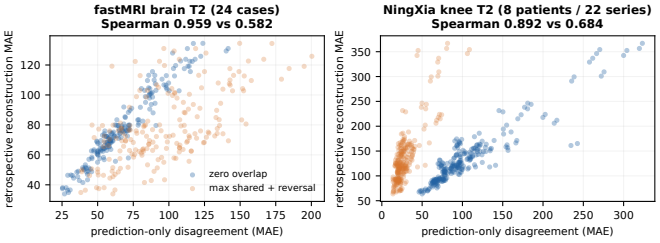


Fig. 3. Frozen all-plane paired intervention. Measurement-disjoint scores rank the same stored target error more strongly than the maximal-shared reversal control. Inference clusters examinations/patients.

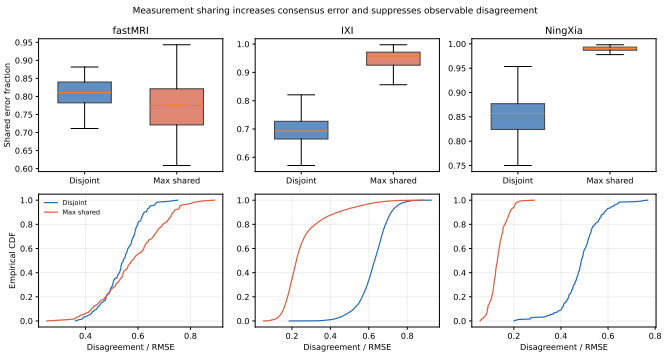


Fig. 4. Consensus distributions and normalized disagreement across contexts. In all 12 strata, more shared error energy predicts less observable disagreement conditional on total error and examination.

B. Shared Error Suppresses Observability

Across every cohort, architecture, and context, shared fraction had a strong negative partial association with RMSE-normalized disagreement after controlling total error and examination: -0.746 to -0.923 on fastMRI, -0.806 to -0.975 on IXI, and -0.894 to -0.959 on NingXia (Fig. 4). Thus suppression was not merely caused by low-error images having small output differences. Equation 2 closed to numerical precision.

Strict stable-wrong prevalence was zero for disjoint contexts. Under maximal sharing it was zero on fastMRI, 0.28–0.32% on IXI, and 2.42–3.23% on NingXia. Patient-clustered NingXia intervals were wide. The failure is coherent but rare under this definition.

The prevalence calculation uses one pooled threshold per dataset/model across both contexts, preventing each condition from defining its own easy quartile. IXI HAT increased by 0.324 percentage points (clustered CI 0.014–0.865); other intervals included zero. NingXia upper limits reached 6.6–8.8%. These estimates reject both extreme interpretations: shared contexts do permit high-error/low-disagreement planes, but the current strict phenotype is not common enough to estimate a clinical incidence.

C. Seven Endpoints and Two Architectures

All 42 dataset–architecture–endpoint point differences favored disjoint contexts (Fig. 5). Architecture-pooled disjoint/shared correlations for MAE were 0.957/0.840 (fastMRI),

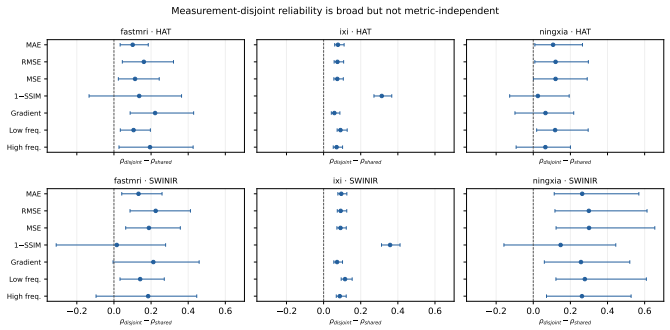


Fig. 5. Seven-endpoint disjoint-minus-shared Spearman differences with examination/patient-clustered intervals. All point estimates are positive; structural/frequency uncertainty is retained.

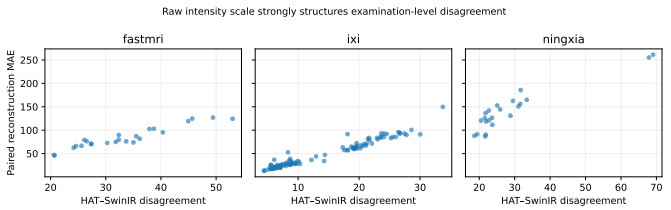


Fig. 6. Intensity operating range. A shared case transform preserves the paired overlap direction but can change cross-examination ranks; independent output transforms are not a valid absolute-discrepancy control.

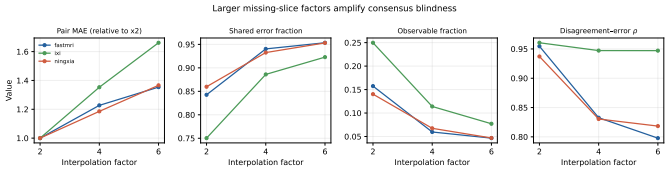


Fig. 7. Analytical factor response. As missing-slice factor increases, error and shared fraction rise while observable fraction and disagreement reliability fall in every cohort.

0.977/0.891 (IXI), and 0.862/0.676 (NingXia). RMSE differences were 0.193/0.083/0.210; low-frequency differences were 0.123/0.103/0.198. The direction persisted separately for HAT and SwinIR.

Structural error is the main boundary. SSIM-error correlations were near zero on fastMRI and negative on NingXia for both contexts, although their disjoint-minus-shared differences remained positive. Only IXI had consistently positive SSIM ranking and intervals excluding zero. Some gradient/high-frequency intervals also crossed zero in smaller strata. Pixelwise disagreement therefore ranks broad intensity error but is not a universal structural-error surrogate.

Table II reports architecture-pooled estimates without endpoint selection. Low-frequency error closely follows MAE, indicating that the result is not driven only by edge noise. The SSIM rows show why direction and absolute utility must be separated: a positive difference can mean "less poor" rather than useful structural-error ranking.

A prespecified Plan7 contrast tested whether blindness was systematically stronger for intensity/low-frequency endpoints than for structural/high-frequency endpoints. It was not: the family contrast was negative for IXI HAT/SwinIR ($-0.068/-0.074$, clustered intervals below zero), positive for NingXia ($0.064/0.063$, one interval crossing zero), and uncertain on fastMRI. Endpoint heterogeneity is therefore cohort dependent; the data support neither metric independence nor a universal frequency-family mechanism.

D. Factor, Geometry, and Intensity Robustness

From $\times 2$ to $\times 6$, mean shared fraction increased $0.843 \rightarrow 0.953$ on fastMRI, $0.750 \rightarrow 0.923$ on IXI, and $0.860 \rightarrow 0.953$ on NingXia (Fig. 7). Reliability decreased $0.955 \rightarrow 0.798$, $0.961 \rightarrow 0.947$, and $0.937 \rightarrow 0.818$, respectively. Paired MAE increased in all cohorts while absolute disagreement generally decreased; IXI had a small $\times 6$ rebound without reliability recovery. The factor phenomenon therefore occurs in transparent analytical estimators and is not exclusively neural.

Across 322 $\times 4$ HAT/SwinIR output volumes, shapes, orientations, and affines were identical. All physical planes matched at every tolerance from 10^{-6} to 10^{-1} mm; maximum affine and plane-distance differences were zero. Applying one shared per-case robust transform to predictions and target preserved a positive disjoint-minus-shared contrast in all six architecture–cohort strata (0.121–0.700). It did not preserve cross-examination ranks, especially on IXI. Independent output normalization changed ranks further and is invalid for absolute discrepancy.

Figure 6 visualizes the scale boundary. A global affine transform with one positive scale would preserve ranks exactly; a per-case scale need not. Applying different scales to each prediction is worse because their difference contains artificial target/offset terms. Consequently, a reliability system must freeze an intensity convention before choosing a triage threshold.

Table III gives the factor summaries. Mean paired MAE rises in every cohort while observable fraction contracts by about threefold. Absolute disagreement falls monotonically on fastMRI and NingXia; IXI changes $48.2 \rightarrow 45.2 \rightarrow 46.6$ without reliability recovery. Thus neither high nor low target error follows mechanically from raw disagreement magnitude.

The geometry audit rules out tolerance-induced membership changes: normal-angle differences were at most 1.48×10^{-6} degrees and physical distance was zero. The intensity audit instead establishes an operating boundary. Raw committee disagreement correlated with examination error by 0.923/0.957/0.874 across the cohorts. A per-case common transform reduced these to 0.784/0.256/0.239. The within-case overlap direction is robust, but between-case triage requires a declared common scale.

E. Selective Risk

Table V and Fig. 8 report complete examination-level curves. Measurement-disjoint nAURC was 0.0119/0.0013/0.0107 on fastMRI/IXI/NingXia, substantially lower than maximal-shared and committee disagreement in every cohort. It was not universally best: adjacent-anchor MAE achieved 0.0070 on NingXia. Disjoint scoring is the

TABLE II
ARCHITECTURE-POOLED SPEARMAN CORRELATIONS FOR EVERY PLAN6 ENDPOINT

Cohort	Endpoint	Disjoint	Shared	Δ	Cohort	Endpoint	Disjoint	Shared	Δ
fastMRI	MAE	.957	.840	.117	IXI	MAE	.977	.891	.086
	RMSE	.936	.743	.193		RMSE	.972	.888	.083
	MSE	.936	.785	.151		MSE	.971	.889	.082
	SSIM error	.006	−.070	.076		SSIM error	.417	.081	.336
	Gradient	.640	.423	.217		Gradient	.973	.908	.065
	Low frequency	.960	.836	.123		Low frequency	.975	.872	.103
	High frequency	.562	.372	.189		High frequency	.969	.891	.079
NingXia	MAE	.862	.676	.185					
	RMSE	.807	.597	.210					
	MSE	.807	.597	.210					
	SSIM error	−.240	−.326	.086					
	Gradient	.629	.468	.161					
	Low frequency	.864	.666	.198					
	High frequency	.642	.479	.164					

TABLE III
ANALYTICAL FACTOR RESPONSE

Cohort	Metric	×2	×4	×6
fastMRI	Pair MAE	112.1	137.6	151.8
	Shared fraction	.843	.940	.953
	Reliability ρ	.955	.833	.798
IXI	Pair MAE	49.8	67.4	82.8
	Shared fraction	.750	.886	.923
	Reliability ρ	.961	.947	.947
NingXia	Pair MAE	157.1	186.2	214.6
	Shared fraction	.860	.933	.953
	Reliability ρ	.937	.830	.818

TABLE IV
HIGH-ERROR EXAMINATIONS RETAINED AT 70% COVERAGE

Score	fastMRI	IXI	NingXia
Measurement-disjoint	0	0	1
Maximal-shared	1	2	1
Architecture committee	0	1	1
Adjacent-anchor MAE	1	3	1

TABLE V
EXAMINATION-LEVEL RELIABILITY (LOWER NAURC IS BETTER)

Score	fastMRI	IXI	NingXia
Measurement-disjoint	.0119	.0013	.0107
Maximal-shared	.0684	.0139	.0306
Architecture committee	.0193	.0140	.0504
Adjacent-anchor MAE	.0212	.0143	**.0070**

most reliable tested output-based construction, whereas a cheaper input statistic can win in an external cohort.

At fixed coverage, disjoint IXI risk was 16.58 at 10% versus oracle 16.55 and was exactly oracle (21.70) at 50%. On fastMRI, several methods selected the same easiest small subset but diverged at intermediate coverage: disjoint 50% risk was 63.65, compared with 65.36 maximal-shared and 65.97 adjacent-anchor. On NingXia, adjacent MAE improved 50% risk from 111.24 to 109.30. No threshold is chosen because clinical capacity and error costs were not specified.

Architecture disagreement is not independent of case difficulty. HAT–SwinIR case-error rank correlations were 0.989/1.000/0.997, and examination identity accounted for 99.1%/99.99%/99.78% of two-architecture squared-error variation. Two architectures can therefore agree about which cases are hard while sharing the dominant error. This explains why committee disagreement is a strong baseline yet remains dependence preserving.

We also counted operational false retention using a threshold frozen as the top quartile of the common examination error and fixed 30/50/70/90% coverages. At 70% coverage, measurement-disjoint scoring retained 0 of 6 high-error fastMRI examinations, compared with 1/0/1 under maximal-shared, committee, and adjacent-anchor scores. On IXI it retained 0 of 29, compared with 2/1/3; the disjoint advantage over maximal-shared and adjacent scores was 0.069 (95% clustered CI 0–0.167) and 0.103 (0–0.188). All NingXia methods retained 1 of 6 at this coverage, with wide patient-clustered intervals. At 90% coverage scores converged because capacity forced most examinations to remain. Thus the operational benefit is concentrated at intermediate coverage and is strongest in IXI; it is not a universal threshold or clinical miss rate.

F. Retrospective Densification Consequence

A score can rank current error without identifying cases that benefit from denser sampling. We therefore evaluated stored $\times 4 \rightarrow \times 2$ improvement as a separate consequence. The original same-family analysis correlated disjoint disagreement with MAE reduction at 0.811 (case-clustered CI 0.592–0.904) on fastMRI and 0.626 (patient-clustered CI 0.200–0.844) on NingXia. At a frozen top-25% budget, selected benefit was 33.36 versus 21.37 under random fastMRI selection and 35.39 versus 24.25 on NingXia.

Plan6 required the permitted IXI+fastMRI pair for confirmation. Before any IXI test output, we froze the lowest-validation-loss eligible I3Net ×2 checkpoint, its SHA-256, the endpoint, and 10/20/30/50% budgets. All 115 IXI outputs and matching ground truths/inputs/metric rows completed. Anchor-CV, adjacent MAE, and output PC correlated with realized benefit by 0.948/0.950/0.970 on IXI and 0.871/0.846/0.843

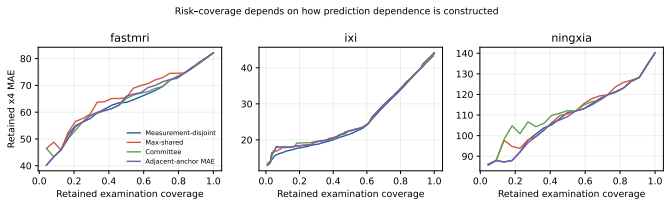


Fig. 8. Risk–coverage against one stored mean HAT/SwinIR MAE endpoint. Disjoint disagreement beats dependence-preserving output scores in all cohorts; adjacent-anchor MAE wins on NingXia.

TABLE VI
FROZEN SCORE CORRELATION WITH X4→X2 BENEFIT

Score	fastMRI	IXI
Anchor-CV	.871	.948
Adjacent-anchor MAE	.846	.950
Output PC	.843	.970

on fastMRI. Every score had positive clustered gain above random at every budget; IXI Anchor-CV captured 0.920–0.973 of oracle gain.

This result supports examination-level consequence, not prospective policy or a unique score. IXI benefit subtracts frozen I3Net $\times 2$ error from mean frozen HAT/SwinIR $\times 4$ error, so it is cross-estimator. The high performance of all three scores is consistent with intrinsic case difficulty. A same-model randomized acquisition study is required to identify sampling treatment benefit.

G. Broad Reconstruction Benchmark

Table VII establishes that the mechanism test is not restricted to two weak or unusually related reconstructors. All entries use the same 24 fastMRI examinations, generated-plane support, ground-truth-derived 1% evaluation mask, and metric implementation. The table is descriptive rather than a leaderboard claim: I3Net is strongest on this benchmark, while CycleINR is less accurate but supplies the most distinct coordinate-based error geometry.

H. Generalization Across Interpolation Paradigms

Figure 9 reports the preregistered physical-dependence intervention. Disjoint contexts increase observable error by 8.0 points for I3Net (95% CI 7.2–8.8), 13.7 for SAINT (12.5–14.9), and 22.6 for CycleINR (21.9–23.3). HAT and SwinIR reference gains are 17.8 and 17.6 points. The equal-weight four-family estimate is 15.5 points [14.8,16.2], and leave-one-family-out estimates remain 13.1–18.0 points with every case-level difference positive. Thus local error geometry, rather than examination ranking, is the robust cross-paradigm endpoint.

Ranking improvement is not universal. Disjoint-minus-shared error-ranking $\Delta\rho$ is .342 [.113,.711] for CycleINR, .018 [−.061,.107] for I3Net, −.010 [−.143,.114] for SAINT, and .057/.062 with intervals crossing zero for HAT/SwinIR. The preregistered stable-wrong rule yields four of 120 shared CycleINR planes and zero for the other models, with zero

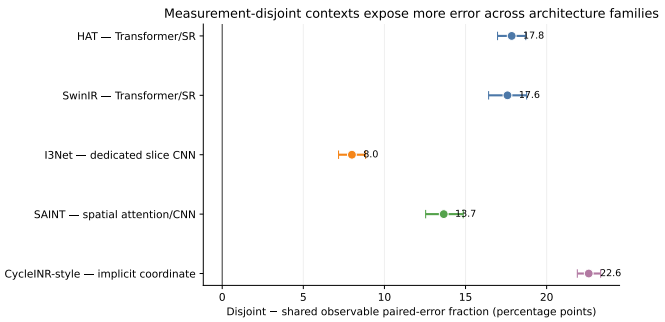


Fig. 9. Architecture-family falsification at fastMRI $\times 4$. Points are disjoint-minus-shared observable paired-error fraction with examination-clustered 95% intervals. No architecture is omitted for an unfavorable ranking result.

disjoint CycleINR events. This is an architecture-specific rare phenotype, not a common prevalence estimate.

Architectural distance also fails to guarantee examination-level independence. Across the five models, median pairwise case-error rank is .945, .981, and .981 at $\times 2/\times 4/\times 6$; six of 24 cases fall in the top error quartile for all five models at $\times 4$ and $\times 6$. Pixel geometry is more diverse: median HAT–SwinIR error cosine is .918, versus .564 for CycleINR–I3Net, with corresponding observable fractions .041 and .245. The data therefore distinguish useful local error diversity from the persistent intrinsic difficulty of entire examinations.

I. Failure Granularity

Examination scores need not localize a failed plane. Historical analysis found 14 NingXia planes in the lowest PC quartile and highest error quartile; 12 were at the first generated $\times 4$–$\times 6$ position, suggesting shared boundary context. Controlled common blur also reduced variation while preserving error. Conversely, a strict natural fastMRI stable-wrong rule found no event. These differences motivate four failure classes: stable-wrong consensus, benign diversity, boundary-sensitive error, and global examination difficulty.

The distinction prevents an examination-level ranking from being presented as a heatmap. A high case score can prioritize review but cannot identify which slice or anatomy is wrong; a median case score can suppress an isolated catastrophic plane. Plane-localization and examination-triage endpoints must be validated separately.

VI. DISCUSSION

The experiments connect a familiar correlated-error limitation to a recorded imaging variable. Through-plane anchors are not generic augmentations: their physical positions determine the anatomical evidence available for the same missing plane. When contexts reuse them, errors align and a larger fraction cancels. Disjoint contexts rank intensity error and selective risk more effectively across three cohorts, while analytical factor response shows worsening consensus blindness as sampling becomes sparser.

The result separates diversity, observability, and calibration. Diversity is output distance; observability is the fraction of paired error surviving in that distance; calibration maps a score

TABLE VII
COMMON-SUPPORT FASTMRI RECONSTRUCTION BENCHMARK. MAE IS REPORTED AT EACH FACTOR; PSNR/SSIM ARE SHOWN AT $\times 4$.

Method	Paradigm	MAE $\times 2$	MAE $\times 4$	MAE $\times 6$	PSNR $\times 4$	SSIM $\times 4$
Linear	Analytical	98.7	134.5	148.7	14.24	.588
Cubic	Analytical	114.5	153.2	148.7	13.21	.544
I3Net	Inter/intra-slice CNN	45.6	70.9	81.6	19.03	.871
SAINT	Spatial conditional CNN	72.1	80.1	90.9	18.30	.848
CycleINR	Implicit coordinate	80.3	114.4	128.2	15.38	.666
HAT	Transformer SR	68.0	86.1	96.7	17.80	.828
SwinIR	Transformer SR	65.7	90.4	97.7	17.37	.803

to an absolute or probabilistic statement. We study the first two. A condition can have larger raw disagreement yet poorer ranking, and no reported number supports statements such as "this slice is correct with a given probability."

Several boundaries change the practical recommendation. First, local observability generalizes more strongly than examination ranking: SAINT's ranking contrast is slightly negative and I3Net's is null within uncertainty. Second, endpoint choice matters: MAE disagreement did not rank SSIM error in two cohorts. Third, intensity scale matters for examination triage even though the paired direction survives a fair common transform. Fourth, stable-wrong consensus remains possible but is rare and model specific under the frozen rule. Fifth, adjacent-anchor MAE beats all output scores on NingXia. Reliability evaluation should record which information changes, compare strong acquired-input baselines, and use the intended harm endpoint.

The literature boundary is also precise. Correlated errors, systematic-error blindness, and leave-one-slice-out evaluation are not new [7], [11], [13]. Recent acquisition perturbations in prostate MRI also show that variance maps can emphasize acquisition fragility while correlating only moderately with error [19]. The defensible contribution here is the target-preserving physical measurement-overlap intervention and its three-cohort endpoint, factor, geometry, failure, and selective-risk characterization.

A. Cohort Heterogeneity

fastMRI supplies the strongest historical all-plane contrast but the smallest public brain panel. Its Plan6 MAE differences are 0.101 for HAT and 0.132 for SwinIR, while SSIM ranking is essentially null. The analytical factor decline is also largest on this cohort. This combination suggests excellent observability for global intensity error but no evidence that the same score orders structural similarity.

IXI supplies the largest and most precise replication. All 14 architecture-specific endpoint intervals exclude zero; the shared-fraction association reaches -0.975 in the maximal-shared HAT stratum, and disjoint risk is nearly oracle. Yet IXI is also most sensitive to per-case normalization. Its evidence supports a robust physical-dependence direction, not a transferable absolute score threshold.

NingXia changes anatomy and provenance and uses out-of-fold models, but eight patients limit inference. Its maximal-shared stable-wrong prevalence is highest, SSIM ranking is negative, and adjacent-anchor MAE wins selective risk. These results argue for retaining simple acquired-input checks in an external knee system and for targeted boundary analysis, not for an anatomy-specific causal claim.

B. Comparator Semantics and Cost

The compared scores vary different information. Disjoint phase changes physical anchor identity and requires two reconstructions plus potentially additional measurements. Maximal-shared reversal changes order but reuses evidence. The committee changes architecture while preserving acquisition. Adjacent-anchor MAE changes no output and requires zero neural calls. Predictive performance must therefore be interpreted with information availability rather than treated as an abstract leaderboard.

Representative Anchor-CV CPU scoring took 0.42/3.97/3.93 ms on fastMRI/IXI/NingXia and stored no predicted volume. Committee, flip-TTA, and disjoint pairs required two neural calls, 0.25–0.29 s / 2.43–3.20 s / 0.49–0.60 s by cohort, about 271 MiB peak allocated GPU memory, and 8.6–66.5 MiB outputs. These measurements establish computational differences only. A disjoint score can be computationally feasible yet acquisition-expensive; a zero-call input baseline can be immediately available. The flip-TTA row is deliberately a cost comparator only: a definition-matched IXI TTA reliability output was unavailable, so no three-cohort nAURC is imputed. Output-PC times are likewise score-only and exclude generating the two prerequisite reconstructions.

C. Clinical Interpretation and Validation

A low disagreement score means only that tested contexts agree under the declared scale. It cannot exclude common smoothing, anatomy omission, or pathology loss. A high score can flag a pair but cannot identify which output or structure is wrong. No score should be displayed as a probability of diagnostic correctness.

The next technical study should freeze this pipeline on an independent multi-site T2 cohort, including boundary and pathology-enriched strata. A prospective paired acquisition should randomize two interleaved phases, acquire a dense reference, quantify inter-phase motion, and use an order-symmetric model when possible. Power must follow patient counts rather than planes. Only after technical replication should a reader study evaluate abstention, review prioritization, lesion detection, quantitative bias, and rescan burden.

The practical recommendation should be falsified if a matched-order disjoint control fails in multiple cohorts, if

TABLE VIII
RELIABILITY OPERATING REGIMES; LATENCY IS MEASURED ON ONE REPRESENTATIVE EXAMINATION PER COHORT

Score	Neural calls	Measured latency	Required information	Recommended regime
Adjacent-anchor MAE	0	0.69/4.00/5.24 ms	Existing sparse anchors	Default screen when no complementary reconstruction exists; strongest tested NingXia nAURC.
Anchor-CV	0	0.42/3.97/3.93 ms	Existing anchors with an interior holdout	Interpretable acquired-only alternate; not uniquely informative.
Output PC	2	1.06/10.20/11.63 ms after inference	Two stored model outputs	Useful only when outputs already exist; generation cost is not zero.
Architecture committee	2	0.25/2.43/0.49 s	Two trained models and outputs	Use when both architectures are already deployed; dependence remains.
Flip TTA	2	0.26/3.18/0.60 s	One model, same anchors, transform	Comparator for transformation variability, not a physical-independence test.
Measurement-disjoint	2	0.29/3.20/0.56 s	Complementary physical phase grids	Worth considering when complementary reconstructions already exist and intermediate-coverage QA matters; otherwise acquisition cost dominates.

gains disappear after motion/intensity control, if a zero-call adjacent baseline consistently matches output scoring, or if clinically important stable-wrong failures remain frequent at the selected threshold. The present study supports that validation program but no deployment decision.

D. Limitations

All phase grids are retrospective simulations from dense T2 volumes; no scanner prospectively changed sampling. NingXia has only eight independent patients. The full three-cohort endpoint matrix remains HAT/SwinIR and retains the historical reversal control's order confound; the clean five-architecture extension is confined to 24 fastMRI examinations and $\times 4$. SAINT and CycleINR are adapted/internal reproductions rather than exact official implementations. The clean in-plane shared control is still an equivariance control, not a prospective acquisition. The analytical factor experiment changes target availability and distance together. MAE/SSIM are not clinical endpoints, and there is no reader study, pathology-stratified analysis, calibrated uncertainty, or prospective adaptive acquisition.

Eligible planes are fully bracketed and exclude acquired anchors, so results need not transfer to unbracketed boundary extrapolation. The 42 endpoint rows are correlated rather than independent replications. Shared fraction is target-dependent and diagnostic, not deployable. Risk–coverage uses one frozen examination aggregation; maximum-plane or reader-task risk could reorder methods. Cross-cohort intensity calibration and prospective scanner-time utility remain unresolved.

E. Evidence and Reproducibility Boundaries

Every Plan6 experiment writes a per-observation table, summary table/JSON manifest, source script, and figure generated from those records. Historical all-plane results and newly balanced three-plane results retain separate file prefixes; neither overwrites the other. The geometry table contains each output identity and affine-derived match. The IXI $\times 2$ record freezes the checkpoint hash before inference and verifies all 115 cases.

The principal journal evidence is the physical dependence contrast, energy distributions, endpoint/factor robustness, architecture falsification, geometry/intensity controls, and selective risk. The journal uses the cross-architecture experiment to characterize MRI QA and does not import the conference paper's detailed finite-response statistic as a new method contribution. Shared cohorts and the architecture experiment must be disclosed under the venue policy; they are not independent studies merely because the framing differs.

The exact ethics and release record remains an author dependency. Public-data terms must be cited from the releases actually used, and internal NingXia approval/waiver language must identify the institution and decision. Until those fields are filled, the manuscript is scientifically compiled but not submission-ready.

Disjoint scoring may require physical measurements unavailable in a conventional sparse scan. Compute cost and acquisition cost differ. A prospective validation should acquire randomized interleaved phases plus a dense reference, quantify motion, predefine a reader-relevant endpoint and coverage threshold, and stop if a matched-order disjoint control fails or a zero-call input baseline consistently matches it.

F. Ethics and Data Availability

fastMRI and IXI are public deidentified resources. NingXia is an internal human MRI cohort. **[AUTHOR ACTION: insert approving/exempting institution, decision identifier, consent/waiver, and permitted-use language.]** Code, manifests, case-level derived tables, and checkpoint hashes will be released anonymously where licensing permits; raw public and internal data remain under their governing terms.

VII. CONCLUSION

Disagreement observes differences between reconstruction errors, not error itself. Across fastMRI, IXI, and patient-clustered NingXia, disjoint physical anchors made intensity-domain error more observable than shared-anchor output comparisons. A preregistered fastMRI falsification extends the local error-geometry effect from transformers to I3Net,

TABLE IX
CLAIM BOUNDARY SUMMARY

Question	Evidence	Conclusion
Does measurement disjointness improve error observability?	All-plane paired intervention; 42 endpoint contrasts; risk curves	Supported for tested deterministic interpolation and intensity-domain endpoints.
Does shared energy suppress normalized disagreement beyond total error?	12 partial rank strata	Supported conditionally; association is not causal mediation.
Is the effect an affine/tolerance artifact?	322 output geometry audit; common-scale control	No for tested geometry and paired direction; cross-case triage remains scale dependent.
Does it generalize over interpolation paradigm?	Five neural architectures at fastMRI $\times 4$; leave-one-family-out analysis	Observable fraction improves in every evaluated architecture; examination-ranking benefit is heterogeneous and not universal.
Does it generalize over factor?	Common-support 13-method benchmark; analytical $\times 2/4/6$ mechanism	Reconstruction error worsens across factors and analytical blindness increases; not a five-architecture neural mechanism test at $\times 2/6$.
Is disjoint disagreement universally best?	Four-score nAURC and false-retention counts	No; best tested output score, adjacent MAE wins NingXia; IXI operational benefit is coverage dependent.
Are stable-wrong failures common?	Strict pooled-quartile prevalence	No; coherent but rare and imprecise in eight-patient NingXia.
Does retrospective benefit imply prospective acquisition utility?	IXI+fastMRI frozen consequence	No; case-level cross-estimator prediction only.

an adapted SAINT-style network, and an internal CycleINR-style model, but also falsifies universal ranking improvement. Different architectures continue to fail on many of the same examinations. The supported principle is therefore narrow: diversify influential physical evidence, validate against the intended endpoint and architecture, and never treat agreement as a correctness certificate.

REFERENCES

[1] C. Peng, W.-A. Lin, H. Liao, R. Chellappa, and S. K. Zhou, "SAINT: Spatially aware interpolation network for medical slice synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 7750–7759. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Peng_SAINT_Spatially_Aware_Interpolation_NeTwork_for_Medical_Slice_Synthesis_CVPR_2020_paper.html

[2] H. Song, X. Mao, J. Yu, Q. Li, and Y. Wang, "I3net: Inter-intra-slice interpolation network for medical slice synthesis," *IEEE Transactions on Medical Imaging*, vol. 43, no. 9, pp. 3306–3318, 2024.

[3] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," *Proceedings of the 33rd International Conference on Machine Learning*, vol. 48, pp. 1050–1059, 2016. [Online]. Available: https://proceedings.mlr.press/v48/gal16.html

[4] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[5] V. Edupuganti, M. Mardani, S. Vasanawala, and J. M. Pauly, "Uncertainty quantification in deep MRI reconstruction," *IEEE Transactions on Medical Imaging*, vol. 40, no. 1, pp. 239–250, 2021.

[6] D. Narnhofer, A. Effland, E. Kobler, K. Hammernik, F. Knoll, and T. Pock, "Bayesian uncertainty estimation of learned variational MRI reconstruction," *IEEE Transactions on Medical Imaging*, vol. 41, no. 2, pp. 279–291, 2022.

[7] S. Hu, N. Pezzotti, and M. Welling, "Learning to predict error for MRI reconstruction," in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021*. Springer, 2021, pp. 604–613.

[8] T. Küstner, K. Hammernik, D. Rueckert, T. Hepp, and S. Gatidis, "Predictive uncertainty in deep learning-based MR image reconstruction using deep ensembles: Evaluation on the fastMRI data set," *Magnetic Resonance in Medicine*, vol. 92, no. 1, pp. 289–302, 2024.

[9] C. de Boor, *A Practical Guide to Splines*. New York: Springer, 1978.

[10] W. Fang, Y. Tang, H. Guo, M. Yuan, T. C. W. Mok, K. Yan, J. Yao, X. Chen, Z. Liu, L. Lu, L. Zhang, and M. Xu, "CycleINR: Cycle implicit neural representation for arbitrary-scale volumetric super-resolution of medical data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 11 631–11 641. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Fang_CycleINR_Cycle_Implicit_Neural_Representation_for_Arbitrary-Scale_Volumetric_Super-Resolution_of_CVPR_2024_paper.html

[11] A. Horváth, S. Pezold, M. Weigel, K. Parmar, and P. C. Cattin, "High order slice interpolation for medical images," in *Simulation and Synthesis in Medical Imaging*. Springer, 2017, pp. 69–78.

[12] R. Tanno, D. E. Worrall, E. Kaden, A. Ghosh, F. Grussu, A. Bizzi, S. N. Sotiropoulos, A. Criminisi, and D. C. Alexander, "Uncertainty modelling in deep learning for safer neuroimage enhancement: Demonstration in diffusion MRI," *NeuroImage*, vol. 225, p. 117366, 2021.

[13] G. Brown, J. Wyatt, R. Harris, and X. Yao, "Diversity creation methods: A survey and categorisation," *Information Fusion*, vol. 6, no. 1, pp. 5–20, 2005.

[14] L. A. Ortega, R. Cabañas, and A. Masegosa, "Diversity and generalization in neural network ensembles," in *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, vol. 151. PMLR, 2022, pp. 11 720–11 743. [Online]. Available: https://proceedings.mlr.press/v151/ortega22a.html

[15] J. Zbontar, F. Knoll, A. Sriram, M. J. Muckley, M. Bruno, A. Defazio, M. Parente, K. J. Geras, J. Katsnelson, H. Chandarana, Z. Zhang, M. Drozdzal, A. Romero, M. Rabbat, P. Vincent, J. Pinkerton, D. Wang, N. Yakubova, E. Owens, C. L. Zitnick, M. P. Recht, D. K. Sodickson, and Y. W. Lui, "fastMRI: An open dataset and benchmarks for accelerated MRI," in *arXiv preprint arXiv:1811.08839*, 2018. [Online]. Available: https://arxiv.org/abs/1811.08839

[16] F. Knoll, J. Zbontar, A. Sriram, M. J. Muckley, M. Bruno, A. Defazio, M. Parente, K. J. Geras, J. Katsnelson, H. Chandarana, Z. Zhang, M. Drozdzal, A. Romero, M. Rabbat, P. Vincent, J. Pinkerton, D. Wang, N. Yakubova, E. Owens, C. L. Zitnick, M. P. Recht, D. K. Sodickson, and Y. W. Lui, "fastMRI: A publicly available raw k-space and DICOM dataset of knee images for accelerated MR image reconstruction using machine learning," *Radiology: Artificial Intelligence*, vol. 2, no. 1, p. e190007, 2020.

[17] Biomedical Image Analysis Group, Imperial College London, "IXI dataset," 2014, accessed: 2026-08-13. [Online]. Available: https://brain-development.org/ixi-dataset/

[18] Y. Ding, J. Liu, J. Xiong, and Y. Shi, "Revisiting the evaluation of uncertainty estimation and its application to explore model complexity-uncertainty trade-off," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pp. 4–5, 2020. [Online]. Available: https://openaccess.thecvf.com/content_CVPRW_2020/html/w1/Ding_Revisiting_the_Evaluation_of_Uncertainty_Estimation_and_Its_Application_to_CVPRW_2020_paper.html

[19] N. M. van der Velde *et al.*, "Aleatoric uncertainty in accelerated prostate MRI reconstruction: Echo-train dropout versus gaussian noise monte carlo sampling," *European Radiology Experimental*, 2026, pMID: 42192056.